

Faridpur Engineering College

1st year 1st semester (CSE)

CT-2, Chem-1104, Chemistry.

Time: 40 minutes

#Each question contain equal 4 mark

Total marks: 20

1. what is K_p and K_c . Explain temperature effect the equilibrium constant for exothermic and endothermic reaction.
2. Drive an equation of K_p for the reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$, where a is the initial concentration of N_2O_4 . α is the degree of dissociation and P is the total pressure.
3. 4mol of HI was placed in 5 liter vessel at $450^\circ C$ at equilibrium mixture found 0.442 mol of I_2 . Calculate the K_c value of the mixture.
4. Define "activated complex state" and explain it with a suitable graph.
5. Drive the rate equation of $A \rightarrow \text{Product}$ reaction.